**Roll No.** | | | | | | | | | | | | | **Total No. of Pages : 02**

**Total No. of Questions : 09**

**B.Tech. (Computer Engineering) (Sem.–5)**

# COMPUTER NETWORKS

**Subject Code : BTCS 504-18**

**M.Code : 91019**

**Date of Examination : 22-12-22**

**Time : 3 Hrs.** **Max. Marks : 60**

**INSTRUCTIONS TO CANDIDATES :**

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

## SECTION-A

**1. Write briefly :**

a) Suggest a suitable LAN topology for a small Laboratory. Justify its choice.

b) Compare FDM and WDM multiplexing techniques.

c) What is virtual LAN? How are connections made in it?

d) What is the role of Firewall? How many types of firewall are there?

e) List the services of Transport layer.

f) What is the need of subnetting? Identify the host and network address of IP 10.05.65.5/21

g) What is Framing? Give its types.

h) Define network Interface and Services of communication model.

i) Explain the role of ARP and RARP protocols and their position in TCP/IP model.

j) Explain SNMP protocol roles and responsibilities.



https://www.ptustudy.com

## SECTION-B

2. Explain the roles and responsibilities of Data Link layer.

3. Give in detail all transmission impairments and way to measure them.

4. Give details of stop & wait ARQ protocol. List its drawback. How are they improved in Go-back-N?

5. State and explain Link State based routing algorithms.

6. Describe possible reasons of congestion in a network. Discuss two variants of leaky bucket for congestion control.

## SECTION-C

7. Give details of OSI model and factors that affected its success.

8. Compare and contrast CSMA/CD and CSMA/CA protocols of MAC.

9. Give the 3 main protocols of Transport layer.

**NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student**.



https://www.ptustudy.com